

# 7B Week 5 Discussion Solutions

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1. We have

$$3.2 \times 10^{-2} \text{N} = \frac{kq_1q_2}{r^2}, \quad (1)$$

which means

$$F_{\text{new}} = \frac{kq_1q_2}{(\frac{1}{8}r)^2}, \quad (2)$$

Therefore,  $F_{\text{new}} = 2.0\text{N}$ .

2.  $F_s = -kx$

(a) In equilibrium,  $\sum \vec{F} = \vec{0}$ . Therefore,

$$\vec{F}_e + \vec{F}_s = 0 \quad (3)$$

$$\frac{1}{4\pi\epsilon_0} \frac{Q^2}{D^2} - k_s(D - L) = 0 \quad (4)$$

(b) Same as (a).

$$(c) -\frac{1}{4\pi\epsilon_0} \frac{Q^2}{D^2} - k_s(D - L) = 0$$

3. last page

4. (a)

$$dF = \frac{1}{4\pi\epsilon_0} \frac{q dQ}{(y + D)^2} \quad (5)$$

$$F = \int_0^L \frac{1}{4\pi\epsilon_0} \frac{q\lambda}{(y + D)^2} dy \quad (6)$$

$$= \frac{q\lambda}{4\pi\epsilon_0} \left( \frac{1}{D} - \frac{1}{D + L} \right) \quad (7)$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q\lambda L}{D(L + D)} \quad (8)$$

$$= \frac{1}{4\pi\epsilon_0} \frac{qQ}{D(L + D)} \quad (9)$$

(b) The forces are equal and opposite as per Newton's third law.

5. (a)

$$Q = \int dq \quad (10)$$

$$= \int_0^L \frac{\lambda_0}{L} y dy \quad (11)$$

$$= \frac{\lambda_0 L}{2} \quad (12)$$

(b)

$$F = \int_0^L \frac{1}{4\pi\epsilon_0} \frac{qy\lambda_0/L}{(y+D)^2} dy \quad (13)$$

$$= \frac{Q\lambda_0}{4\pi\epsilon_0 L} \int_0^{L/D} \frac{x}{(1+x)^2} dx; \text{ where } x = y/D \quad (14)$$

$$= \frac{Q\lambda_0}{4\pi\epsilon_0 L} \left( \frac{1}{L/D+1} + \ln(L/D+1) - 1 \right) \quad (15)$$

$$= \frac{Q\lambda_0}{4\pi\epsilon_0 L} (\ln(1+L/D) - \frac{L}{d+L}) \quad (16)$$

6.

$$F = \left( -\frac{Qq}{4\pi\epsilon_0} \frac{1}{(d-s)^2} + \frac{Qq}{4\pi\epsilon_0} \frac{1}{(d+s)^2} \right) \hat{x} \quad (17)$$

$$= \frac{Qq}{4\pi\epsilon_0} \frac{1}{d^2} \hat{x} \left( -\frac{1}{(1-s/d)^2} + \frac{1}{(1+s/d)^2} \right) \quad (18)$$

$$\approx \frac{Qq}{4\pi\epsilon_0} \frac{1}{d^2} \hat{x} (-(1+2s/d) + (1-2s/d)) \quad (19)$$

$$= -\frac{Qq}{\pi\epsilon_0} \frac{s}{d^3} \hat{x} \quad (20)$$

So, we see that we have a force pushing us back to equilibrium, proportional to  $s$ . This means (similar to spring oscillation) that the period of oscillation is:

$$T = 2\pi \sqrt{\frac{m}{\frac{Qq}{\pi\epsilon_0} \frac{1}{d^3}}} = 2\pi \sqrt{\frac{\pi\epsilon_0 d^3 m}{Qq}} \quad (21)$$

3.

From the diagram, we see that the  $x$  components of the two fields will cancel each other at the point P. Thus the net electric field will be in the negative  $y$ -direction, and will be twice the  $y$ -component of either electric field vector.

$$\begin{aligned}
 E_{\text{net}} &= 2E \sin \theta = 2 \frac{kQ}{x^2 + a^2} \sin \theta \\
 &= \frac{2kQ}{x^2 + a^2} \frac{a}{(x^2 + a^2)^{1/2}} \\
 &= \boxed{\frac{2kQa}{(x^2 + a^2)^{3/2}} \text{ in the negative } y \text{ direction}}
 \end{aligned}$$

